Semester: 5th
Subject Name:- DMDW& Code:- IT-3031
Branch (s): - CSE, IT

AUTUMN MID SEMESTER EXAMINATION-2023

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Data Mining and Data Warehousing
[IT-3031]

Time: 1 1/2 Hours **Full Mark: 40**

Answer Any four Questions including Question No. 1 which is compulsory.
The figures in the margin indicate full marks. Candidates are required to give theiranswers in their own words as far as practicable and all parts of a question should beanswered at one place only.

1. Answer all the questions. [2 x 5]

   a) Explain the term ETL with respect to Data Warehouse ?

   b) Explain Stratifies sampling with example.

   c) Differentiate between OLAP and OLTP.

   d) Compare Drill Down with Roll Up approach.

   e) Explain the term Variance and Standard Deviation with example.

2. (a) What are the steps involved in KDD process ? Explain each step in detail. [5 Marks]

   (b) Find the Euclidean distance matrix & Manhatan distance matrix of below points. [5 Marks]

Point	X	Y
P1	0	2
P2	5	0
P3	4	3
P4	6	2

3. (a) What is data transformation? Explain the data transformation using **z-score** normalisation for the data points {10,15, 20, 50, 60}. [5 Marks]

   (b) What is data Warehouse? Explain Fact Constellation schema for multidimensional datamodel with example. [5 Marks]

4. (a) What is Bessel's correction? Calculate and draw a Box plot for a dataset {35, 42, 38, 45, 48, 60, 55, 40, 50, 53, 41, 36}. [5 Marks]

   (b) What is IQR? Determine if there are any potential outliers in the dataset given in Q.4.(a) using the IQR method. If there are outliers, identify them and provide their values. [5 Marks]

5. (a) What is the Apriori property? The transaction data is given in the table below from a restaurant. There are 9 distinct orders (Order:1 – Order:9) and each transaction involves between 2 and 4 meal items. There are a total of 5 meal items that are involved in the orders. For all of the parts below the minimum support is 2/9 and the minimum confidence is 6/9. Find all strong association rules of the form: $X \wedge Y \rightarrow Z$ and note their confidence values.

[10 Marks]

Meal Item	List of Item IDs	Meal Item	List of Item IDs
Order:1	M1,M2,M5	Order:6	M2,M3
Order:2	M2,M4	Order:7	M1,M3
Order:3	M2,M3	Order:8	M1,M2,M3,M5
Order:4	M1,M2,M4	Order:9	M1,M2,M3
Order:5	M1,M3		

*** Best of Luck ***